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MATKUL : ELEKTRONIKA KOMUNIKASI

1 Sebuah rangkaian resonansi paralel terdiri dari induktor 50nH, faktor kualitas 85 dan kapasitor 20pF, faktor kualitasnya 100. Tentukan frekuensi resonansi, resonansi paralel dan bandwidth rangkaian resonansi tersebut dihubungkan dengan sumber dan beban dengan RL = Rs = 1000Ω, berapa faktor kualitas dan bandwidth rangkaian semarang?

Jawab :

• Diket : L = 50nH, Q = 85, C = 20pF, Q = 100
• Ditt : F, Rparalel dan BW? Q dan BW → dihubungkan ke sumber

» $F_0 = \frac{1}{2\pi\sqrt{LC}}$

$= \frac{1}{2\pi\sqrt{5\times10^{-8}\cdot2\cdot10^{-8}}}$

$= \frac{1}{6{,}28\cdot\sqrt{1\times10^{-8}}}$

$= \frac{1}{6.28\times10^{-9}} = 159{,}235\,MHz$

» $W_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{50\cdot10^{-9}\times20\cdot10^{-12}}}$

» $R = W_0 \cdot L \cdot Q_V$
$= 10^9 \times 5\times10^{-8}\times 85 = 4{,}25\,K\Omega$

» $FBW = \frac{1}{Q_V} = \frac{1}{85} = 0{,}011176$

» $BW = F_0 \times FBW$
$= 159{,}235\,MHz \times 0{,}011176 = 1{,}872\,MHz$

» $R_{ext} = 1000 \,//\, 1000 = 500\Omega$

$Q_{ext} = \frac{R_{ext}}{W_0 \cdot L} = \frac{500}{10^9\cdot15\cdot10^{-8}} = 10$

$Q_L = \frac{Q_a \cdot Q_{ext}}{Q_V \cdot Q_{ext}} = \frac{85\,(10)}{85+10} = 8{,}947$

$BW = \frac{F_0}{Q_L} = \frac{159{,}235\,MHz}{8{,}947} = 17{,}797\,MHz.$

1.2 Rancanglah sebuah rangkaian resonansi paralel dengan F = 100 MHz dan BW = 10MHz. Induktor yang digunakan memiliki faktor kualitas kapasitor 100. Diket RL = 2Rs. berapa nila resistansi sen dari induktor dan kapasitor.

↳ Jawab :

$R_{ext} = R_1 \,//\, 2R_S = 0{,}6 \times R_S \rightarrow R_{ext} = \frac{R_S \cdot 2R_S}{R_S + 2R_S} = \frac{2R_S^2}{3R_S} = \frac{2}{3}R_S$

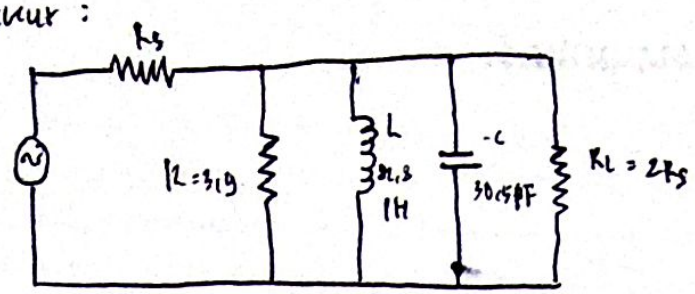
$Q_L = \frac{F}{BW} = \frac{100\,MHz}{10\,MHz} = 10$

$Q_{ext} = \frac{Q_V \cdot Q_L}{Q_u \cdot Q_L} = \frac{75\cdot10}{75\cdot10} = 11{,}53$

$L = \frac{R_{ext}}{W_0 \cdot Q_{ext}} = \frac{0{,}6}{2\pi\cdot10^8\cdot11{,}53} = 82{,}8\,pH$, $C = \frac{Q_{ext}}{R_{ext}\cdot W_0} = \frac{11{,}53}{0{,}62\times10^9} = 30{,}15\,nF$

$K = W_0 \cdot L \cdot Q_u$
$= 2\pi\cdot10^8\cdot82{,}8\cdot10^{-12}\cdot75$
$= 3{,}9\Omega$

▶ Jadi resistansi seri dan induktor kapasitor adalah 319 Ω dengan rangkaian sebagai berikut :



MATCHING IMPEDANCE

2.1 Sebuah beban dengan impedansi $Z_L = (60 - j20)\,\Omega$ pada $f = 100$ MHz akan disesuaikan ke sumber dengan impedansi $Z_S = 50\,\Omega$ menggunakan matching network impact elemen topologi L. Hitung nilai elemen-elemen dari 2 solusi matching network tersebut menggunakan metoda perhitungan (bukan smithchart)

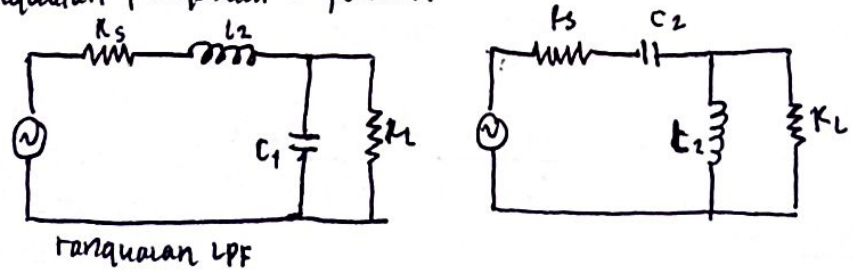
▶ Jawab :

$$X_2 = -X_S \pm R_S B$$
$$= -(0) \pm (50)(0{,}5577)$$
$$= \pm 28{,}85 \longrightarrow +28{,}85 \text{ (induktif)}$$
$$-28{,}85 \text{ (kapasitif)}$$

$$B_1 = \frac{X_L \pm R_L B}{R_L^2 + X_L^2}$$
$$= \frac{(-20) \pm (60)(0{,}5577)}{(60)^2 + (-20)^2} = \frac{14{,}62}{4000} = 3{,}65 \cdot 10^{-3} \text{ (kapasitif)}$$

$$B_1 = \frac{X_L \pm R_L B}{R_L^2 + X_L^2}$$
$$= \frac{(-20) - (60)(0{,}5577)}{(60^2) + (-20^2)} = \frac{-54{,}62}{4000} = -0{,}01365 \text{ (induktif)}$$

▶ Rangkaian penyesuai impedansi



Rangkaian LPF

- Normalisasi Z_L dan Z_S terhadap impedansi $Z_0 = 50\,\Omega$, dimana $Z_0 = Z_S$

$$\hat{Z_L} = \frac{Z_L}{Z_0} = \frac{30 - j20}{50\,\Omega} = 0{,}6 - j0{,}4$$
$$\hat{Z_S} = \frac{Z_S}{Z_0} = \frac{50}{50} = 1$$

- Bergerak dengan lingkaran r konstan, hingga memotong lingkaran faktor kualitas didapatkan potongan $Z_A = 0{,}6 - j$[illegible]

maka $\hat{Z_A} - \hat{Z_L} = 0{,}6 - j3 - (0{,}6 - j0{,}4)$
$= -j3 + j0{,}4$
$= -j2{,}6$

atau $\hat{Z_A} = n - jnL\,\theta$
$= 0{,}6 - j0{,}6 \cdot 5$
$= 0{,}6 - j3$ (sama)

• Dinormalisasi $x_1 \rightarrow x_1 = x \cdot Z_0$

$= 2{,}16 \times 50$

$= 130\Omega$

• $C_1 = x_1 = \frac{1}{\omega C}$

$C_1 = \frac{1}{\omega x_1} = \frac{1}{2\pi \cdot 10^8 \cdot 130\Omega} = 12{,}24 pF$

⇒ Admitansi $Y_A = \hat{2A}$ diubah menjadi Y_A

$Y_A = \frac{1}{\hat{z_A}} = \frac{1}{0{,}6 - j3} = 0{,}064 + j0{,}32$

⇒ bergerak dari Y_A sepanjang garis $Y_A = 0{,}064$

↳ $Y_b = 0{,}064 - j0{,}26$

$Y_A = 0{,}064 + j0{,}32$

$j_B = -j0{,}58$

⇒ $L_1 = \frac{1}{\omega_B} = \frac{1}{2\pi \cdot 10 \cdot 0{,}116} = 137{,}2 nH$

⇒ denormalisasi

$B_2 = \frac{j0{,}58}{50} = 0{,}0116$

⇒ denormalisasi

$x_2 = x_2 \times 50$

$= 3{,}6 \times 50 = 180\Omega$

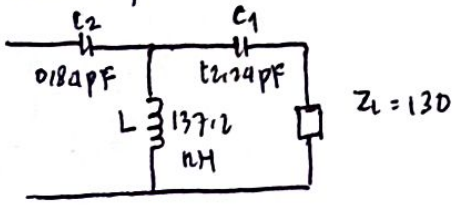
⇒ Y_B diubah menjadi Z_B

↳ $Z_B = \frac{1}{Y_B} = \frac{1}{0{,}064 + j0{,}26} = 0{,}189 + j3{,}6 = 1 + j3{,}6$

⇒ Maka $Z_S - Z_B = 1 - (1 + j3{,}6)$

$x_2 = -j3{,}6$

⇒ Rangkaian topologi T



FILTER

3.1 Low pass filter. Rancanglah sebuah low pass filter LC dengan F cutoff = 100 MHz pada F = 125 MHz diinginkan redaman minimum 20 dB. Gunakan respon filter chebyshev dengan ripple 0,1 dB. Komponen pertama prototipe berupa induktor $R_0 = 50\Omega$. Gambarkanlah rangkaian prototipe dan rangkaian sebenarnya.

Jawab :

⇒ $\Omega_S = \frac{\omega_S}{\omega_C} = \frac{125}{100} = 1{,}25$

⇒ $n \geq \frac{\cosh^{-1}\sqrt{\frac{10^{0{,}1} - 20}{10^{0{,}1} \cdot 0{,}1 - 1}}}{\cosh^{-1} 1{,}25}$

$n \geq \frac{\cosh^{-1} 65{,}19}{\cosh^{-1} 1{,}25} = 7$

$n = 7$

⇒ Nilai elemen - elemen

$L_1 = \frac{R_0 \cdot g}{\omega C} = \frac{50 \cdot 1{,}1812}{2\pi \cdot 100M} = 94 nH$

$C_2 = \frac{g}{R_0 \cdot \omega C} = \frac{1{,}4228}{50 \cdot 2\pi \cdot 100M} = 45 pF$

$L_3 = \frac{50 \cdot 2{,}0967}{2\pi \cdot 100M} = 167 nH$

$C_4 = \frac{1{,}5734}{50 \cdot 2\pi \cdot 100M} = 50 pF$

$L_5 = L_3 = 167 nH$ $\quad L_7 = L_1 = 94 nH$

$C_6 = C_2 = 45 pF$

» Didapatkan rangkaian dengan :

$n = 7$

$g_1 = g_7 = 1{,}1812$

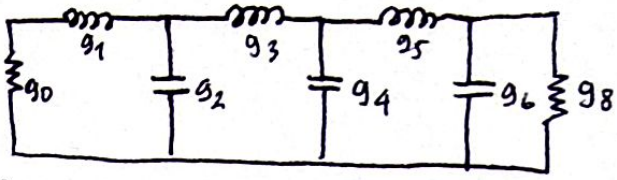
$g_2 = g_6 = 1{,}4228$

$g_3 = g_5 = 2{,}0967$

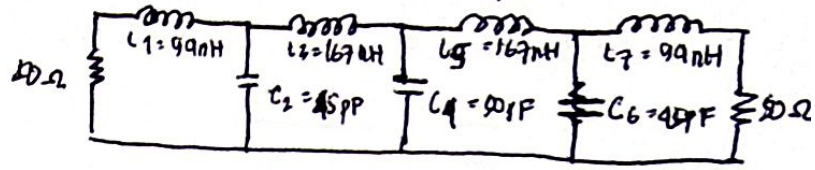
$g_4 = 1{,}57254$

$g_8 = 1$

ⓐ Gambar rangkaian prototipe



ⓑ Gambar rangkaian sebenarnya



3.2 Bandpass filter : rancanglah sebuah bandpass filter dengan F tengah 100 MHz dan Bandwidth 10 MHz. Pada BW = ±50 MHz. Redaman minimum 60 dB dan $R_0 = 50\Omega$. Gunakan respons Chebyshev ripple 0,17 dB Gambarkan rangkaian prototipe dan rangkaian sebenarnya.

Jawab :

» $FBW = \dfrac{W_2 - W_1}{W_0} = \dfrac{10}{100} = 0{,}1$

» Mencari transformasi frekuensi pada frekuensi 50 MHz

$\Omega_s = \dfrac{1}{FBW} * \left(\dfrac{W}{W_0} - \dfrac{W_0}{W}\right) = \dfrac{1}{0{,}1}\left(\dfrac{50}{100} - \dfrac{100}{50}\right)$

$= 10\,(0{,}5 - 2) = 10\,(-1{,}5) = -15$ (ambil nilai mutlaknya)

» Menentukan ordo filter :

$n \geq \dfrac{\cosh^{-1}\sqrt{\dfrac{10^{2}-1}{10^{0{,}01}-1}}}{\cosh^{-1} 15} \geq 6{,}29$

$n \geq \dfrac{9{,}410763}{3{,}400084} \geq 6{,}29 \Rightarrow n = 2{,}788 \Rightarrow n = 3.$

↳ untuk F = 150

$\Omega = \dfrac{1}{0{,}1}\left(\dfrac{150}{100} - \dfrac{100}{150}\right)$

$\Omega = 8{,}3$

» Agar hasil simetris maka ordo filter menjadi 3

$g_0 = g_4 = 1{,}000$; $g_1 = 1{,}031 = g_3$; $g_4 = 1{,}147$

» Harga elemen-elemen :

(diperoleh dengan elemen pertama berupa L seri)

ⓐ komponen seri

$L_{1P} = \dfrac{R_0 \cdot g}{FBW \cdot W_0} = \dfrac{50 \times 1{,}031}{0{,}1 \times 10^{8}} = 51{,}55\ nH$

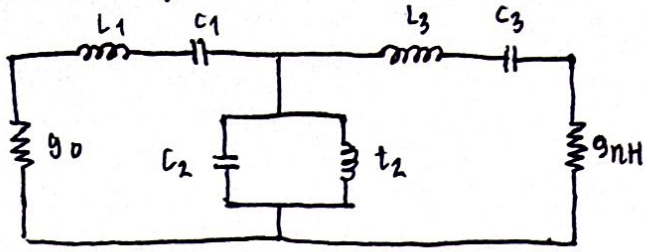
» $C_{S1} = \dfrac{FBW}{R_0 \cdot W_0 \cdot g_1} = \dfrac{0{,}1}{50 \cdot 10^{8} \cdot 1{,}036} = 19{,}38\ pF$

ⓑ komponen paralel

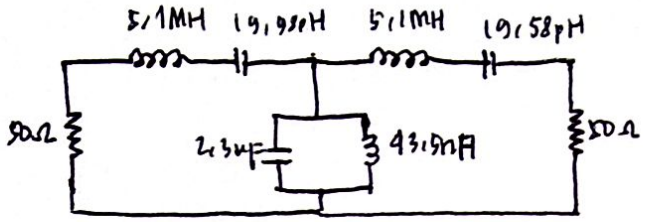
$C = \dfrac{g}{FBW \times W_0 \cdot R_0} = \dfrac{1{,}147}{0{,}1 \times 10^{8} \times 50} = 2{,}294\ nF$

» $L_P = \dfrac{R_0 \cdot FBW}{W_0 B_2} = \dfrac{50 \times 0{,}1}{100 \times 1{,}1479} = 43{,}5\ nH$

➡ Gambar rangkaian prototipe BPF



➡ Gambar rangkaian sebenarnya



NAME	TITLE	DWG. NO.
RAISYA AULIYA SALSABILA	JAWABAN UTS	
SMITH CHART ENGS 120	COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997	DATE 12 April 2020

NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

